

The aim of this section is to develop an understanding of the fundamental principles of heredity, which are illustrated by reference to model organisms and to the inheritance of genetic diseases.

The concept of epigenetics will also be considered in relation to the increase in diseases such as Type 2 diabetes.

Some of the practical techniques used to analyse and manipulate DNA are also considered, as well as the applications of these techniques. The potential uses of DNA and RNA technology in the treatment of disease are considered, as is the profound ethical implications of gene technologies and the role of genetic counsellors.

This section provides opportunities for the use of statistical tests in analysing data.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.

